

Does a Reproducible Promotive/Adversity Structure Exist in HBCD Infancy Data?

HBCD's dense, longitudinal sampling lets us test reproducibility of factor structure across time

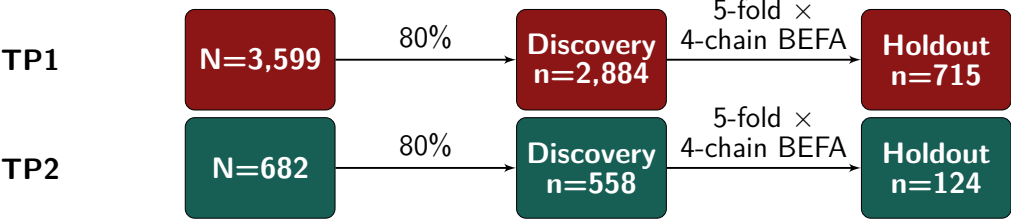
- Conceptual grounding: DMAP (dimensional model of adversity and psychopathology; threat vs. deprivation), unpredictability, ecological systems theory (Bronfenbrenner), promotive vs. protective factors.
- Preregistered battery returned mostly null results.** Extended post-hoc with additional HBCD-collected measures (see Measures table) – disclosed as a deviation.

Q1 (H1): does a low-dimensional promotive/adversity structure exist in HBCD's item battery, and does it reproduce across independent subsets of the same cohort?

Q2 (H2/H3): if so, do promotive-dominant dimensions predict more favorable outcomes, and do adversity-dominant dimensions protectively moderate that association?

Two-stage design: discover, then confirm

- TP1 (prenatal-9mo) and TP2 (9-17mo) modeled independently via Bayesian EFA (BEFA). Rolling enrollment: TP2 N far smaller.



- 18 candidate items at TP1, 24 at TP2 (23 active – full list below).

Measures, visits, and timepoint assembly

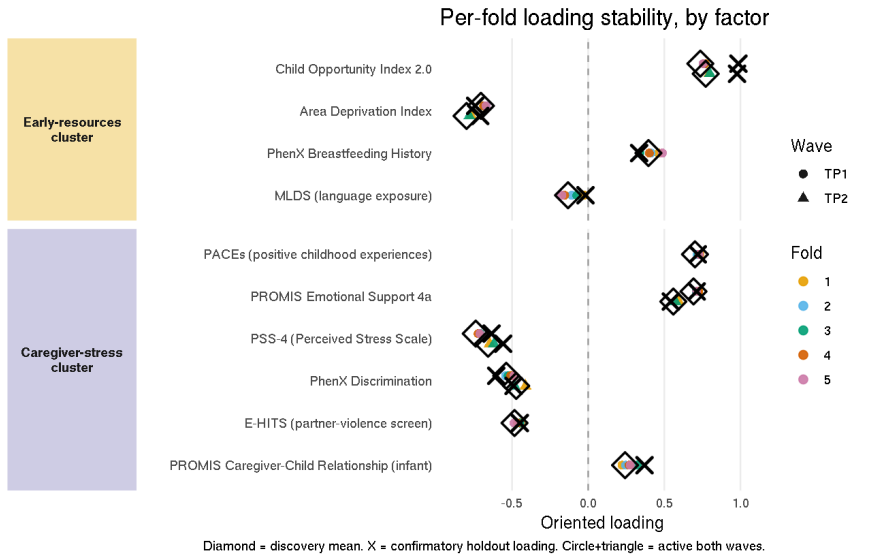
Measure	TP1			TP2	
	V01	V02	V03	V04	V05
Breastfeeding Hist. ^{†*}	–	2255	64	9	–
Area Depriv. Index (ADI) [†]	2649	–	–	–	577
Child Opp. Index 2.0 (COI) [†]	2665	–	–	–	580
MLDS (lang. exposure) [†]	–	–	1284	636	–
Walkability Index [†]	2193	–	–	–	467
Tree canopy cover [†]	2665	–	–	–	580
Neighborhood safety [†]	3500	12	–	–	378
Econ. hardship: eviction [†]	3502	14	–	–	382
Econ. hardship: general [†]	3051	12	–	–	343
Food insecurity [†]	–	2219	1312	627	–
Transitions-in-Care [†]	–	1057	743	521	52
E-HITS	3395	12	–	–	–
PhenX Discrimination [†]	3378	11	–	–	371
ecPROMIS CG-Child Rel. [†]	–	–	1323	–	28
PACEs (current)	3473	10	–	–	–
PROMIS ES4a [†]	3507	2202	1302	629	–
PSS-4 [†]	3499	2203	1303	630	–
IBQ-R (child temperament) [†]	–	–	1308	–	352
ScreenQ (screen time)	–	–	–	641	–
HOME Inv.: Cog. Stimulation	–	–	–	615	–
PACEs (retrospective)	–	–	–	–	382
HOME Inv.: Emotional Support	–	–	–	615	–
CASR-SF (partner abuse)	–	–	–	530	–
PEARLS (pediatric ACEs)	–	–	–	601	–
ACEs (adult)	–	–	–	588	–
ecPROMIS CG-Child Rel. [†]	–	–	–	–	392
CHAOS (unreflected)	–	–	–	634	–
ECBQ (child temperament)	–	–	–	–	74

[†] shared across timepoints (same instrument or GLED/PROMIS parallel form). * excluded at TP2, too sparse (<10% coverage). Outcomes: 3 Bayley (cognitive/motor/ language) subscales, CDI-WG (vocabulary), Vineland-ABC (adaptive behavior), PROMIS-Peer (peer relations), SPM-2 (sensory processing), MAP-DB (disruptive behavior). Covariates: site, maternal education/income, age.

No consistent factor structure for environmental variables

Timepoint	Factors (M)	Folds agreeing	CFI (holdout)	RMSEA
TP1	4	3/5 – FAIL	0.903 (scaled)	0.048
TP2	4	1/5 – FAIL	0.798	0.107

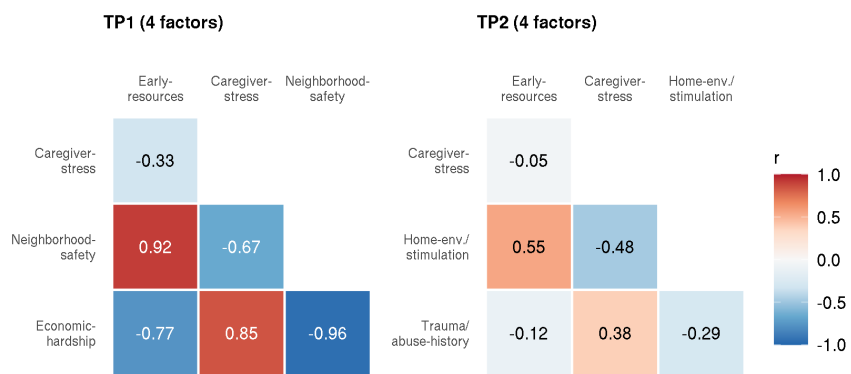
- Held-out fit: TP1 fails CFI (0.903 vs. conventional threshold of 0.95) but clears RMSEA (0.048 vs. conventional threshold of 0.06); TP2 fails both (0.798, 0.107).
- Adding IBQ-R, GLED-TP2, and CHAOS post-preregistration changes item grouping but not instability.



Factor membership diverges between timepoints

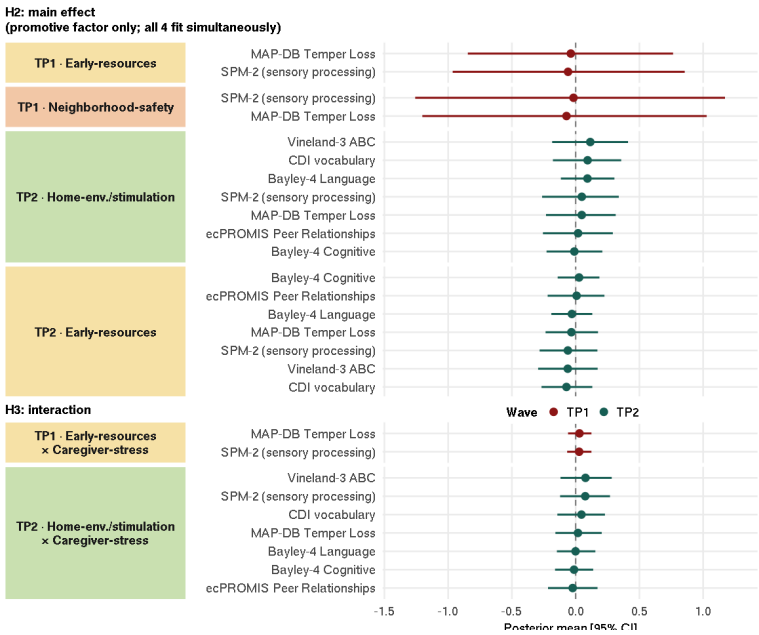
	TP1	TP2
<i>Early-resources</i>	COI+, ADI–, Breastfeeding+, MLDS– (n.s.)	COI+, ADI–
<i>Caregiver-stress</i>	PACEs+, ES4a+, Discrimination–, E-HITS–, ecPROMIS+, IBQ-R–	PSS-4–, Discrimination–, CHAOS–, ES4a+
<i>Nbhd.-safety</i>	Safety+, Eviction–, Tree canopy+	HOME-ES+, HOME-CS+, PACEs(retro)+, Hardship–, ScreenQ–, Safety (n.s.)
<i>Econ.-hardship</i>	Hardship+, Food insecurity+	PEARLS+, CASR-SF+, ACEs(adult)+

Signs = holdout loading direction (not magnitude); TP2 realigned to TP1's orientation where factors recur (ADI/COI, PSS-4/ES4a/Discrimination), since BEFA's sign is arbitrary per timepoint. n.s. = not significant.



Absence of evidence for promotive/adversity–outcome effects

- H2:** outcome ~ 4 factors + age + site, simultaneous.
- H3:** H2 + each timepoint's clearest promotive × adversity interaction.



TP1's Early-resources factor predicts fewer sensory/temper problems in marginal model

Marginal (one-factor-at-a-time) models – drop every other factor covariate, keep age/site only:



- The signal:** tested alone, this factor predicts fewer sensory-processing (SPM-2, -0.155) and temper-loss (MAP-DB, -0.151) problems.
- Confounds explain only part of the signal:** TP1's only 2 outcomes are both caregiver-reported on nearly the same sample ($r = 0.437$).
- Cross-timepoint check:** shared items are adversity-heavy, model fit actually worse across chains and folds.
- SES check:** partialling out maternal ed./income (+ site) doesn't shrink ADI/COI's correlation (-0.69 to -0.73) – but shrinks breastfeeding/ language exposure's link to outcomes near zero.

Conclusions & future directions

- Neither timepoint reproduced – but ADI/COI's correlation stayed stable ($r = -0.69$ TP1, -0.70 TP2).
- Absence of evidence for H2/H3 association.
- TP1's factor collinearity limits independent signal from any one factor.
- Food for thought:** neighborhood factors merge ($|r|$ up to 0.96), yet ADI/COI stay distinct from household items (SES check) – nested (Bronfenbrenner) levels? Next: CFA, nested vs. single-factor, on holdout.
- Next step:** cross-timepoint measurement-invariance test on future data to disambiguate developmental change vs. instrument-switching.